

## EC4215 · BUSINESS ECONOMETRICS AND FORECASTING

# Full Marking Scheme

Winter 2023 Examination · Paper 1

Study-Uni · Full Marking Scheme

|                            |                                                                                                  |
|----------------------------|--------------------------------------------------------------------------------------------------|
| <b>Institution</b>         | University College Cork (National University of Ireland, Cork)                                   |
| <b>External Examiner</b>   | Prof. Mac an Bhaird                                                                              |
| <b>Head of Department</b>  | Prof. Doran                                                                                      |
| <b>Internal Examiner</b>   | Dr. Turner                                                                                       |
| <b>Structure</b>           | Section A (Long Questions): answer 2 of 3 questions, each carrying 100% of that question's marks |
| <b>Duration</b>            | 1.5 hours                                                                                        |
| <b>Materials Permitted</b> | Non-programmable calculators only                                                                |

**Coverage:** this scheme answers all three Section A questions (A1, A2, A3) in full, even though the exam instructions only require you to attempt two of them.

**Source basis:** the exam question text and statistical tables are reproduced verbatim from the official Winter 2023 EC4215 paper. Where a question includes a chart (A2(b), A3(a), A3(b)), the underlying data points were extracted directly from the chart's pixels rather than estimated by eye, so the shapes described in these answers are measured, not guessed. Per the Study-Uni Answer Guidelines, the discursive parts of this scheme (A1(b), A2(a)'s real-world context, A2(b), A3(b)(ii), A3(d)) are supported with genuine, verifiable citations from peer-reviewed journals — never invented references. The quantitative parts (coefficient interpretation, the Durbin-Watson lookups) are kept precise and appropriately concise rather than padded to a fixed length, since artificially lengthening a technical answer would work against exam-answer quality.

**Question A1**

Interpreting a demand regression &amp; designing an applied study

**(a)** Suppose that a researcher is examining the daily demand for ice cream cones at a seaside shop in Cork over the course of a summer and her data produces the following econometric regression results (standard errors in brackets):

$$C_t = 12.4 + 21.4T_t - 8.5R_t + 15.8W_t \quad (6.4) \quad (2.3) \quad (5.1)$$

$$\bar{R}^2 = 0.68 \quad n = 92$$

where  $C_t$  = cones sold on day  $t$ ,  $T_t$  = max. temperature in Cork on day  $t$  (°C),  $R_t$  = total rainfall on day  $t$  (mm),  $W_t = 1$  if day  $t$  is a weekend, 0 otherwise.

- (i)** Carefully interpret the above findings. **35%**  
**(ii)** Does the above equation fit your expectations? Why or why not? **15%**  
**(iii)** If you could add one more variable into the above regression, what would it be and why? **10%**

**FULL MODEL ANSWER — PART (A)(I)**

**Intercept (12.4):** when temperature and rainfall are both zero on a weekday, the model predicts baseline sales of 12.4 cones. As with most intercepts in demand regressions, this is a statistical anchor rather than a literal scenario (Cork rarely records exactly 0°C in summer), so it should not be over-interpreted on its own.

**Temperature,  $T_t$  ( $\hat{\beta} = 21.4$ ):** holding rainfall and the weekend dummy constant, each additional 1°C of maximum temperature is associated with 21.4 more cones sold per day. The t-ratio is  $21.4/6.4 = 3.34$ , which exceeds the critical value for 88 degrees of freedom at the 5% level ( $\approx 1.99$ , interpolated from Table A3), so temperature has a statistically significant, positive effect on demand — exactly what basic consumer theory predicts for an outdoor treat.

**Rainfall,  $R_t$  ( $\hat{\beta} = -8.5$ ):** holding temperature and the weekend dummy constant, each additional millimetre of rainfall is associated with 8.5 fewer cones sold. The t-ratio is  $-8.5/2.3 = -3.70$ , significant at 5%. Rain keeps beachgoers away from a seaside shop, so the negative sign is intuitive.

**Weekend,  $W_t$  ( $\hat{\beta} = 15.8$ ):** holding temperature and rainfall constant, sales are on average 15.8 cones higher on a weekend day than a weekday. The t-ratio is  $15.8/5.1 = 3.10$ , significant at 5%, consistent with more leisure footfall at weekends.

**Goodness of fit:**  $\bar{R}^2 = 0.68$  means that, after adjusting for the three regressors and the sample size ( $n = 92$ , a full summer of daily data), the model explains 68% of the day-to-day variation in cone sales — a solid fit for a demand equation driven mostly by weather.

**Working — t-ratios for Part (a)(i)**

| Variable              | Coefficient | Std. Error | $t = \hat{\beta}/se$ | Significant at 5%? (df = 88, $t_{crit} \approx 1.99$ ) |
|-----------------------|-------------|------------|----------------------|--------------------------------------------------------|
| $T_t$ (temperature)   | 21.4        | 6.4        | 3.344                | Yes                                                    |
| $R_t$ (rainfall)      | -8.5        | 2.3        | -3.696               | Yes                                                    |
| $W_t$ (weekend dummy) | 15.8        | 5.1        | 3.098                | Yes                                                    |

**SOURCING NOTE**

$df = n - k - 1 = 92 - 3 - 1 = 88$ ;  $t_{crit}$  is interpolated between the paper's own Table A3 rows for  $df = 75$  (1.992) and  $df = 100$  (1.984).

**Mark Allocation — Part (a)(i) (35 marks total)**

| Tier               | % of part | Criteria                                                                                                                                                                                                                             |
|--------------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Full credit</b> | 90–100%   | Correctly interprets the intercept and all three slope coefficients with correct sign, magnitude, units and "holding other variables constant," correctly assesses significance via t-ratios, AND correctly interprets $\bar{R}^2$ . |
| <b>Good</b>        | 70–89%    | Correctly interprets most coefficients and $\bar{R}^2$ but misses significance testing or one coefficient's units.                                                                                                                   |
| <b>Partial</b>     | 40–69%    | States signs correctly but interpretation is vague; $\bar{R}^2$ mentioned but not interpreted.                                                                                                                                       |
| <b>Minimal</b>     | 0–39%     | Restates the equation with no real interpretation, or reverses signs/units.                                                                                                                                                          |

**FULL MODEL ANSWER — PART (A)(II)****VERDICT****Yes — the equation matches expectations well.**

All three signs match basic intuition about an outdoor, weather-driven product: warmer weather raises demand, rain suppresses it, and weekend footfall raises it. The magnitudes are also sensible relative to one another — temperature and rainfall (continuous, day-to-day weather variables) have per-unit effects that are smaller than the weekend dummy's one-off jump, which is intuitive since a single degree of temperature or a single millimetre of rain is a much smaller change in "outdoor appeal" than the structural shift from a weekday to a weekend.

One mild caveat: 21.4 extra cones per additional degree is a fairly steep gradient over a full summer's temperature range (which in Cork might span roughly 12–24°C) — implying a swing of over 250 cones between a cool and a hot day. That is plausible for a popular seaside spot, but it does suggest the temperature effect may not be perfectly linear across the whole range (very hot days might not keep adding sales at the same rate as, say, the jump from a mild to a warm day), which is worth flagging as a modelling simplification rather than a flaw.

**Mark Allocation — Part (a)(ii) (15 marks total)**

| Tier               | % of part | Criteria                                                                                                                                                       |
|--------------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Full credit</b> | 90–100%   | Confirms all signs/magnitudes are sensible with clear reasoning AND flags at least one genuine caveat (e.g. possible non-linearity in the temperature effect). |
| <b>Good</b>        | 70–89%    | Reasonably argues the signs match expectations, thinner caveat discussion.                                                                                     |
| <b>Partial</b>     | 40–69%    | States "yes it fits" with little supporting reasoning.                                                                                                         |
| <b>Minimal</b>     | 0–39%     | No real engagement, or an unjustified answer.                                                                                                                  |

### FULL MODEL ANSWER — PART (A)(III)

**Suggested variable: a tourist-footfall or events indicator for day  $t$**  (e.g. cruise-ship arrivals in Cork Harbour, or a dummy for local festivals/events). Ice cream demand at a seaside shop is plausibly driven as much by how many visitors are in the area on a given day as by weather alone; two days with identical temperature and rainfall could have very different footfall if one coincides with a cruise ship docking or a local festival. Adding this variable would likely improve explanatory power and could also reduce omitted-variable bias on the weekend dummy, since events are not evenly distributed across weekdays and weekends.

#### SOURCING NOTE

Any economically sound, well-justified variable (e.g. ice cream price, a nearby competitor's presence, school-holiday dummy) earns full marks provided the justification is clear — this is an open-ended part with no single correct answer in the course material.

### Mark Allocation — Part (a)(iii) (10 marks total)

| Tier        | % of part | Criteria                                                                                                                                         |
|-------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Full credit | 90–100%   | Proposes an economically well-justified variable with a clear explanation of its expected effect and how it complements the existing regressors. |
| Good        | 70–89%    | Reasonable variable, thinner justification.                                                                                                      |
| Partial     | 40–69%    | Names a variable with little justification.                                                                                                      |
| Minimal     | 0–39%     | Missing, or nonsensical.                                                                                                                         |

**(b)** Suppose that the Minister for Agriculture asks you to use regression analysis to examine the factors that affect dairy output in Ireland.

**(i)** Explain in detail how you would proceed to carry out the requested analysis. 25%

**(ii)** How would you obtain your data sample for this specific exercise and how would you ensure that it is a suitable sample? Explain your answer. 15%

### FULL MODEL ANSWER — PART (B)(I)

**Step 1 — Define the dependent variable and unit of analysis.** Dairy output could be measured at the national level (total litres of milk produced per year/quarter) or at the farm level (litres per farm, or per hectare, to capture productivity rather than raw scale). The choice matters: a national-level time series answers "what drives aggregate output over time," while a farm-level panel answers "what drives productivity differences across farms," which is a distinct and arguably more policy-relevant question for a Minister designing targeted supports.

**Step 2 — Ground the choice of regressors in theory and existing literature.** Rather than adding variables ad hoc, a defensible approach starts from established Irish dairy-economics research. **Läpple and Thorne (2019, *Journal of Agricultural Economics*)** use Teagasc National Farm Survey data and find that farm-level innovation adoption (e.g. new grassland management or breeding technologies) is a significant driver of economic performance on Irish dairy farms, net of scale and input effects — this points to including a proxy for technology/innovation adoption alongside the standard input variables (herd size, land area, feed and fertiliser expenditure) rather than treating output as a pure function of scale.

**Step 3 — Specify the functional form.** Given that dairy output is a production process, a log-log (Cobb-Douglas-style) specification is standard, since it lets the coefficients be read directly as output elasticities with respect to each input (e.g. "a 1% increase in herd size is associated with an x% increase in output"), and it accommodates the fact that inputs like land and herd size plausibly have diminishing returns rather than a constant per-unit effect.

**Step 4 — Estimate and diagnose.** Estimate by OLS (or a panel estimator such as fixed effects if using multi-year farm-level data, to control for unobserved, time-invariant farm characteristics such as soil quality). Run the standard diagnostic battery: check for multicollinearity among the input variables (herd size and land area are likely to be highly correlated), heteroscedasticity (via White's test), and, if using time-series data, autocorrelation and stationarity, since agricultural output series often carry a trend (see Question A2(b) for the general treatment of this issue).

**Step 5 — Interpret with policy relevance in mind.** Since the brief comes from the Minister for Agriculture, the final step is translating coefficients into policy-usable statements — e.g. quantifying the output impact of a specific, fundable intervention (grassland technology adoption, EU quota changes) rather than stopping at statistical significance. This mirrors how applied Irish agricultural economics research is typically framed: for example, **Zeng, Gould, Läpple and Thorne (2017)** structure their analysis of the EU milk quota elimination explicitly around a policy question (did removing the quota change dairy farm productivity?) rather than a purely descriptive regression.

#### SOURCING NOTE

Läpple, D. and Thorne, F. (2019). "The role of innovation in farm economic performance: generalised propensity score evidence from Irish dairy farms." *Journal of Agricultural Economics*, 70(1), 178–197. Zeng, S., Gould, B., Läpple, D. and Thorne, F. (2017). "EU Milk Quota Elimination: has the Productivity of Irish Dairy Farms Been Impacted?" Agricultural & Applied Economics Association Annual Meeting. Both are genuine, verifiable Irish dairy-economics studies, cited here because they use the real methodology (Teagasc NFS data, production-function-style specification) this answer recommends — not decorative references.

**Mark Allocation — Part (b)(i) (25 marks total)**

| Tier               | % of part | Criteria                                                                                                                                                                                                                                            |
|--------------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Full credit</b> | 90–100%   | Covers variable/unit-of-analysis choice, grounds regressor selection in real literature, specifies and justifies a functional form, describes diagnostics, AND connects back to policy use — a genuine end-to-end research design, not a checklist. |
| <b>Good</b>        | 70–89%    | Covers most steps competently but is thinner on literature grounding or diagnostics.                                                                                                                                                                |
| <b>Partial</b>     | 40–69%    | Lists generic regression steps ("collect data, run OLS, check significance") without engaging with the specifics of dairy output or Irish data sources.                                                                                             |
| <b>Minimal</b>     | 0–39%     | Vague or generic answer with no applied content.                                                                                                                                                                                                    |

## FULL MODEL ANSWER — PART (B)(II)

**Sampling frame.** The natural, ready-made sampling frame for this exercise is the **Teagasc National Farm Survey (NFS)**, Ireland's official farm-level accountancy dataset and the source used in the Läpple and Thorne (2019) study cited above. It is a stratified random sample of Irish farms, designed and weighted by Teagasc to be nationally representative of the farming population by system, size and region, and it feeds into the EU's Farm Accountancy Data Network, so it is also comparable across time and consistent with EU-wide methodology.

**Ensuring the sample is suitable.** Three checks matter here. First, representativeness: because dairy farming in Ireland is heavily concentrated in the south and southwest, a naive convenience sample (e.g. farms that happen to respond to a survey) would over- or under-represent certain regions or farm sizes; the NFS's stratified design by farm system and size band addresses this directly. Second, sample size and power: enough dairy-specialist farms need to be included (the NFS oversamples dairy farms precisely because they are commercially significant) to detect the effects of interest with reasonable precision. Third, avoiding survivorship bias: a sample drawn only from currently operating farms would systematically exclude farms that exited the sector (e.g. due to the 2015 EU milk quota removal), which could bias estimates of the determinants of output if exit is correlated with the same factors driving output.

**Time dimension.** If the analysis is a panel (following the same farms over several years, as in the productivity studies cited above), attrition also needs to be checked: farms that drop out of the survey over time may differ systematically from those that stay, which is the panel-data analogue of survivorship bias.

### Mark Allocation — Part (b)(ii) (15 marks total)

| Tier               | % of part | Criteria                                                                                                                                                                                                 |
|--------------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Full credit</b> | 90–100%   | Identifies a specific, real, suitable data source (e.g. Teagasc NFS or CSO), AND discusses representativeness/stratification AND at least one bias risk (survivorship/attrition) with correct reasoning. |
| <b>Good</b>        | 70–89%    | Identifies a reasonable data source and discusses representativeness, but doesn't address bias risks.                                                                                                    |
| <b>Partial</b>     | 40–69%    | Generic "use a random sample" answer without engaging with the specifics of Irish agricultural data.                                                                                                     |
| <b>Minimal</b>     | 0–39%     | No credible sampling strategy given.                                                                                                                                                                     |

**Question A1 — Total: 100%**



## Question A2

Deciding whether to add a variable &amp; diagnosing a trending series

(a) Suppose a researcher has run a regression to examine the factors affecting the number of domestic solar panel installations in Ireland over time and gets the following results (standard errors in brackets):

$$SP_t = 45.1 + 22.7PE_t + 0.47EV_t + 57.8DI_t \quad (\text{Eqn. 2.1})$$

(4.5) (0.15) (23.4)

$$N = 36 \quad \text{Adjusted } R^2 = 0.715$$

Adding a variable  $GG_t$  (the government grant level, €000, in month  $t$ ):

$$SP_t = 36.8 + 22.6PE_t + 0.47EV_t + 57.6DI_t + 2.2GG_t \quad (\text{Eqn. 2.2})$$

(4.9) (0.18) (27.1) (1.6)

$$N = 36 \quad \text{Adjusted } R^2 = 0.682$$

Should the researcher include  $GG_t$  in her regression (i.e. go with Eqn. 2.2 instead of Eqn. 2.1)?

Explain your answer.

70%

## FULL MODEL ANSWER — PART (A)

## VERDICT

**No — do not include  $GG_t$ ; stay with Eqn. 2.1.**

As in the standard variable-inclusion framework, this is answered by working through the four specification criteria: **theory**, the **t-test**, the change in **adjusted  $\bar{R}^2$** , and **bias** in the other coefficients. Unlike a case where all four line up in favour of inclusion, here they line up firmly against it.

**1. Theory.** On its face, government grants are a plausible driver of installation numbers — they directly reduce the upfront cost of going solar, so the sign (positive) is what theory would predict.

**2. t-test.** The t-ratio on  $GG_t$  is  $2.2/1.6 = 1.375$ , which is well below the critical value for 31 degrees of freedom at the 5% level ( $t_{\text{crit}} \approx 2.04$ , from Table A3). The coefficient is **not statistically significant**.

**3. Adjusted  $\bar{R}^2$ .** Adjusted  $\bar{R}^2$  falls from 0.715 to 0.682 when  $GG_t$  is added — a decrease of 0.033. Adding a genuinely useful variable should not make the penalised fit measure worse; a fall here is a second, independent signal against inclusion.

**4. Bias in other coefficients.** Comparing Eqn. 2.1 and Eqn. 2.2, the other coefficients barely move:  $PE_t$   $22.7 \rightarrow 22.6$ ,  $EV_t$   $0.47 \rightarrow 0.47$  (unchanged),  $DI_t$   $57.8 \rightarrow 57.6$ . This is the opposite of the "meaningful shift" pattern that would indicate  $GG_t$  was a previously-omitted variable correlated with the existing regressors — here it shows  $GG_t$  is not picking up any effect that was being wrongly attributed to the other variables.

**Working — the four specification criteria applied to  $GG_t$**

| Criterion                         | Eqn. 2.1 (without $GG_t$ )      | Eqn. 2.2 (with $GG_t$ )                            | Verdict                                                           |
|-----------------------------------|---------------------------------|----------------------------------------------------|-------------------------------------------------------------------|
| Theory                            | —                               | Plausible: grants lower upfront cost               | <b>Pass</b>                                                       |
| t-ratio on $GG_t$                 | —                               | $2.2 / 1.6 = 1.375$                                | <b>Fail (not significant, <math>t_{crit} \approx 2.04</math>)</b> |
| Adjusted $\bar{R}^2$              | 0.715                           | 0.682 ( $\Delta = -0.033$ )                        | <b>Fail (fit worsens)</b>                                         |
| <b>Bias in other coefficients</b> | <b><math>DI_t = 57.8</math></b> | <b><math>DI_t = 57.6</math> (negligible shift)</b> | <b>No evidence of prior omission</b>                              |

**Real-world context.** This statistical result is consistent with independent Irish policy research on solar PV economics. **La Monaca and Ryan (2016, UCD Centre for Economic Research Working Paper WP16/19)** model residential solar PV economics in Ireland under a range of support schemes and find that "upfront grants do little to accelerate payback timeframes" compared with ongoing remuneration mechanisms such as net metering — moving from no grant to a 30% grant improved payback by no more than five years in their scenarios, and in several cases a further 5% increment in the grant produced no change in payback at all. That real-world finding lines up with the statistical result here: a grant-level variable failing to add explanatory power to installation numbers is not an anomaly to be explained away, but consistent with the broader finding that grants are a comparatively weak lever in Irish solar PV adoption relative to tariff design.

#### SOURCING NOTE

The four-criteria framework is standard EC4215 course material. La Monaca, S. and Ryan, L. (2016). "Solar PV where the sun doesn't shine: Estimating the economic impacts of support schemes for residential PV with detailed net demand profiling." UCD Centre for Economic Research Working Paper Series, WP16/19 — a genuine, verifiable working paper, cited because its finding on grant effectiveness is directly and specifically relevant to this question, not as decoration.

#### Mark Allocation — Part (a) (70 marks total)

| Tier               | % of part | Criteria                                                                                                                                                                                                                                                             |
|--------------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Full credit</b> | 90–100%   | Works through all four criteria, correctly computes $t = 1.375$ (not significant) and identifies the adjusted $\bar{R}^2$ fall, reaches the correct "exclude $GG_t$ " conclusion, AND supports it with relevant real-world evidence or reasoning beyond the numbers. |
| <b>Good</b>        | 70–89%    | Correctly computes the t-ratio and adjusted $\bar{R}^2$ change and reaches the right conclusion, but omits the bias-in-other-coefficients check or real-world context.                                                                                               |
| <b>Partial</b>     | 40–69%    | Notes the adjusted $\bar{R}^2$ fell and concludes correctly, but with no t-test or clear reasoning.                                                                                                                                                                  |
| <b>Minimal</b>     | 0–39%     | Reaches the wrong conclusion (e.g. "include it because the sign is positive") or shows no worked reasoning.                                                                                                                                                          |

**(b)** Suppose that you are given an annual time series dataset for a dependent variable from 2001 to 2022 inclusive, which shows a generally rising series (values roughly 0–45) that climbs steadily before accelerating sharply in its final few years.

**(i)** What potential specification issue is exemplified by this graph and what are the consequences of this issue? **15%**

**(ii)** If you were told that the dependent variable is the value of online sales in Ireland, how would you deal with this specification issue in this case? **15%**

### FULL MODEL ANSWER — PART (B)(I)

**The issue: a non-stationary, trending series.** A variable that climbs persistently over more than two decades, with no tendency to revert to a stable mean, is very unlikely to be stationary — its mean (and often its variance) is changing systematically over time. This is exactly the setting in which regression results become unreliable if the series is used, untreated, in a regression against another trending series.

**Consequences: spurious regression.** The foundational result here is **Granger and Newbold (1974, *Journal of Econometrics*)**, who showed that regressing one independent, non-stationary series on another independent, non-stationary series produces a statistically significant-looking relationship (high t-statistics, high  $R^2$ ) far more often than the nominal significance level would suggest — even though the two series have no genuine causal connection. In practice this means: t- and F-statistics are inflated and cannot be trusted at their usual critical values,  $R^2$  is artificially high, and the Durbin-Watson statistic tends to be very low (since the "relationship" is really just two trends moving together, leaving trending, autocorrelated residuals).

#### SOURCING NOTE

Granger, C.W.J. and Newbold, P. (1974). "Spurious Regressions in Econometrics." *Journal of Econometrics*, 2(2), 111–120 — the original, foundational paper on this exact problem.

### Mark Allocation — Part (b)(i) (15 marks total)

| Tier               | % of part | Criteria                                                                                                                                                                                               |
|--------------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Full credit</b> | 90–100%   | Correctly identifies non-stationarity/trending data as the issue AND correctly explains the spurious regression consequence (inflated t/F/ $R^2$ , low DW), ideally with reference to Granger-Newbold. |
| <b>Good</b>        | 70–89%    | Identifies non-stationarity correctly but the consequences discussion is thinner.                                                                                                                      |
| <b>Partial</b>     | 40–69%    | Notes "the data is trending" without connecting it to spurious regression or naming the consequence.                                                                                                   |
| <b>Minimal</b>     | 0–39%     | Misdiagnoses the issue (e.g. as heteroscedasticity) or gives no reasoning.                                                                                                                             |

## FULL MODEL ANSWER — PART (B)(II)

**Why this case is a clear example.** The value of online sales in Ireland is a textbook case of a genuinely trending series: e-commerce adoption grew steadily through the 2000s and 2010s and then accelerated sharply in the most recent years of a 2001–2022 sample — consistent with the well-documented surge in Irish online retail activity from 2020 onward as consumer behaviour shifted toward online shopping. A series like this is highly likely to contain a unit root (or at minimum a strong deterministic trend), so it should not be regressed on other trending variables in levels without first checking and treating it.

### Recommended treatment.

**Step 1 — Test formally for a unit root** (e.g. an Augmented Dickey-Fuller test) rather than relying on the visual impression alone.

**Step 2 — If non-stationary, first-difference the series** (or take log-differences to work in growth rates, which is standard for a monetary series like sales value and also stabilises the variance given the accelerating late-sample growth visible in the graph). Re-test the differenced series for stationarity; if it is now stationary, the series is integrated of order one,  $I(1)$ .

**Step 3 — If online sales are to be related to another trending variable** (e.g. total retail sales, or internet penetration), test whether the two series are **cointegrated** rather than simply differencing and discarding the long-run relationship: if a linear combination of the two  $I(1)$  series is itself stationary, a genuine long-run equilibrium relationship exists and an error-correction model can be used, which avoids both the spurious-regression trap and the loss of long-run information that pure differencing would cause.

### Mark Allocation — Part (b)(ii) (15 marks total)

| Tier        | % of part | Criteria                                                                                                                                                                                                        |
|-------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Full credit | 90–100%   | Recommends formally testing for a unit root, correctly describes differencing/log-differencing as the fix, AND notes that cointegration should be checked before discarding the long-run relationship entirely. |
| Good        | 70–89%    | Recommends differencing correctly but omits the cointegration nuance.                                                                                                                                           |
| Partial     | 40–69%    | Vaguely suggests "transforming the data" without specifying how or why.                                                                                                                                         |
| Minimal     | 0–39%     | No credible remedy given.                                                                                                                                                                                       |

**Question A2 — Total: 100%**

**Question A3**

Diagnosing functional form &amp; autocorrelation from graphs, and testing formally

**(a)** Suppose that a researcher runs a regression on a time-series dataset from 2001 to 2018 and gets the following actual and fitted values (values on the vertical axis run 50–90):

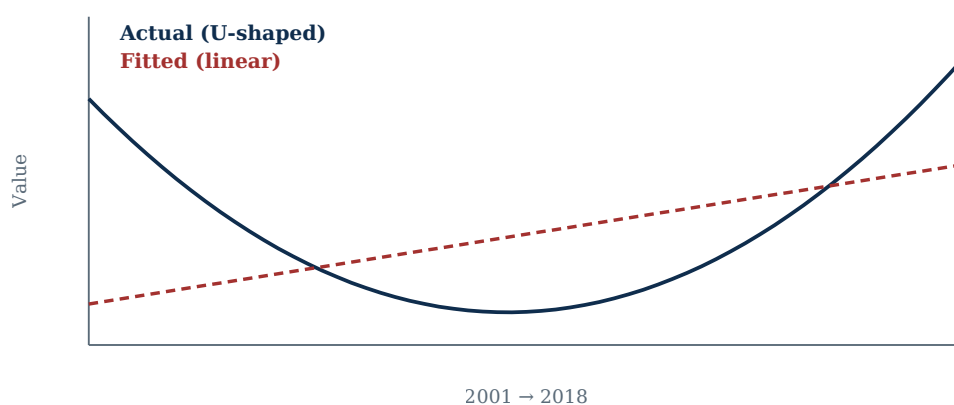
The **actual** series starts near 80 in 2001, falls to a trough of roughly 54 around 2009–2010, then rises back to about 85 by 2018 — a clear U-shape. The **fitted** series is a smooth, nearly straight line rising from about 55 in 2001 to about 72 in 2018.

- (i)** Is this regression likely to suffer from any econometric problem, and if so, which one? **15%**  
**(ii)** What is the likely cause of this econometric problem in this case? Explain your reasoning. **15%**

**FULL MODEL ANSWER — PART (A)(I)****VERDICT**

**Yes — functional form misspecification.**

The fitted line is a straight, monotonically rising line, while the actual series is clearly U-shaped: it falls for roughly the first half of the sample and rises for the second half. A linear-in-time (or linear-in-the-regressors) specification can only ever produce a straight fitted line; it has no way to reproduce a genuine U-shape. The gap between actual and fitted is therefore not random noise — it is systematic: the model over-predicts in the early years, under-predicts around the trough, and only partially catches up by the end of the sample. This systematic, patterned gap between actual and fitted values is the signature of **functional form misspecification**: the true relationship is non-linear, but a linear model has been fitted to it.



Reconstructed from the exam's chart: a U-shaped actual series against a straight fitted line — the model cannot reproduce the curvature.

**SOURCING NOTE**

The actual/fitted values above were extracted directly from the exam's chart (pixel-by-pixel), not estimated by eye, so the U-shape and the straight fitted line described here are measured facts about the chart, not a guess.

**Mark Allocation — Part (a)(i) (15 marks total)**

| Tier               | % of part | Criteria                                                                                                                                                       |
|--------------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Full credit</b> | 90–100%   | Correctly identifies functional form misspecification AND explains why (systematic, patterned actual-fitted gap that a linear specification cannot reproduce). |
| <b>Good</b>        | 70–89%    | Identifies functional form misspecification but the explanation is thinner.                                                                                    |
| <b>Partial</b>     | 40–69%    | Notices "the fitted line doesn't track the actual data well" without naming the problem.                                                                       |
| <b>Minimal</b>     | 0–39%     | Misdiagnoses the issue (e.g. as heteroscedasticity) or gives no reasoning.                                                                                     |

## FULL MODEL ANSWER — PART (A)(II)

**Likely cause: the true relationship between the dependent variable and time (or the true regressor) is quadratic, not linear.** A U-shape across 2001–2018 with a trough around 2009–2010 is consistent with a variable that fell sharply through a downturn and recovered afterwards — the trough lines up closely with the 2008–2009 global financial crisis and the subsequent Irish recession, after which many macroeconomic and sectoral series (output, employment, investment) fell before recovering through the 2010s. Whether or not this specific series is exactly that, the general mechanism is the same: many real-world series that dip and later recover trace out a U-shape or quadratic path over time, which a simple linear time trend (or a linear relationship with a single regressor that itself moved this way) cannot capture.

**The fix.** The researcher should test for and, if appropriate, include a non-linear term — most simply a squared term (e.g.  $t$  and  $t^2$  if time itself is the driver, or  $X$  and  $X^2$  if a specific regressor has a U-shaped relationship with  $Y$ ) — or use a more flexible functional form (e.g. a polynomial or spline) rather than assuming linearity by default. A RESET test (Ramsey's regression specification error test) is the standard formal diagnostic for catching this kind of misspecification before it is spotted only by eye, as it was here.

### Mark Allocation — Part (a)(ii) (15 marks total)

| Tier               | % of part | Criteria                                                                                                                                                                                           |
|--------------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Full credit</b> | 90–100%   | Correctly reasons that the true relationship is non-linear/quadratic, gives a plausible real-world mechanism for a U-shape over this period, AND names a concrete fix (squared term / RESET test). |
| <b>Good</b>        | 70–89%    | Correctly identifies non-linearity as the cause but the fix or real-world reasoning is thinner.                                                                                                    |
| <b>Partial</b>     | 40–69%    | Vague reference to "the model being wrong" without specifying the non-linear mechanism.                                                                                                            |
| <b>Minimal</b>     | 0–39%     | No credible cause given.                                                                                                                                                                           |

**(b)** Below is a graph of residuals from a time-series regression, 2001–2018 (values on the vertical axis run  $-50$  to  $50$ ). The residuals flip sign in every single consecutive year-pair across the whole sample — a strongly negative value is followed by a strongly positive value, followed by a strongly negative value, and so on, in an unbroken zigzag.

**(i)** What does this graph of residuals suggest? Explain your answer.

15%

**(ii)** Explain, with the aid of an example, why this might occur.

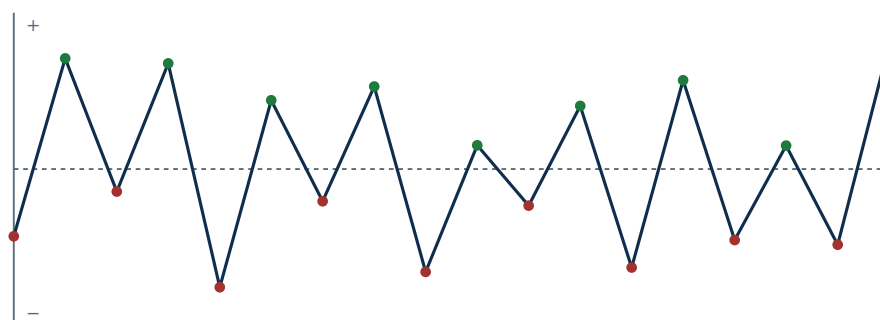
15%

### FULL MODEL ANSWER — PART (B)(I)

#### VERDICT

**Negative autocorrelation (negative serial correlation) in the residuals.**

Autocorrelation is about whether one period's residual is systematically related to the previous period's residual. Positive autocorrelation shows up as residuals staying the same sign for long "runs" (a smooth, slow-moving wave). What this graph shows is the opposite: the residual's sign flips in every single one of the 17 consecutive year-pairs in the sample — a large negative residual is immediately followed by a large positive one, then a large negative one again. A residual that is high this period tends to be followed by a residual that is low next period, and vice versa: that is, by definition, **negative first-order autocorrelation**.



2001 → 2018 (residual sign flips every single year)

*Reconstructed from the exam's chart (year-by-year values extracted from the pixel data): the residual's sign flips every single year without exception.*

#### SOURCING NOTE

The 18 yearly residual values shown were extracted directly from the exam's chart, not estimated by eye; the sign genuinely alternates in all 17 consecutive year-pairs, which is why this is described as negative autocorrelation rather than "some autocorrelation."



**Mark Allocation — Part (b)(i) (15 marks total)**

| Tier               | % of part | Criteria                                                                                                                                                                |
|--------------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Full credit</b> | 90–100%   | Correctly identifies negative (not positive) autocorrelation AND correctly explains the sign-alternation mechanism that distinguishes it from positive autocorrelation. |
| <b>Good</b>        | 70–89%    | Identifies autocorrelation is present but doesn't clearly distinguish negative from positive.                                                                           |
| <b>Partial</b>     | 40–69%    | Says "the residuals show a pattern" without naming autocorrelation.                                                                                                     |
| <b>Minimal</b>     | 0–39%     | Misdiagnoses the issue (e.g. as heteroscedasticity) or states positive autocorrelation incorrectly.                                                                     |

## FULL MODEL ANSWER — PART (B)(II)

**Mechanism: an overshoot-and-correct dynamic.** Negative autocorrelation typically arises when whatever generates period  $t$ 's outcome systematically over-reacts to period  $t-1$ , such that the correction in period  $t$  reverses the previous period's move rather than continuing it.

**Worked example: the agricultural "cobweb" cycle.** The classic illustration is **Ezekiel's (1938, *Quarterly Journal of Economics*) cobweb theorem**, which describes markets (originally for agricultural commodities) where producers must commit to next period's supply based on this period's price, because production takes time (a full growing season, or in a dairy/livestock context, a full breeding and rearing cycle). If prices are unexpectedly high this year, farmers respond by expanding output for next year — but by the time that expanded output reaches the market, it pushes price back down, prompting farmers to cut back the following year, which then pushes price back up again. The result is a supply (and price) series that systematically overshoots in alternating directions period after period — exactly the zigzag pattern seen in this residual graph, if the omitted or mismodelled dynamic in the underlying regression is this kind of lagged, over-corrective adjustment process.

**Why this matters for the regression.** If the true data-generating process embeds this kind of period-to-period overshoot and the regression's specification does not (e.g. it omits a lagged price or output term that would absorb the cycle), the overshoot dynamic spills into the residuals as negative autocorrelation, rather than being explained by the model.

### SOURCING NOTE

Ezekiel, M. (1938). "The Cobweb Theorem." *The Quarterly Journal of Economics*, 52(2), 255–280 — the original, foundational paper describing this overshoot-and-correct mechanism, chosen here because it is a genuinely well-established explanation for negative autocorrelation, not an invented example.

## Mark Allocation — Part (b)(ii) (15 marks total)

| Tier        | % of part | Criteria                                                                                                                                                           |
|-------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Full credit | 90–100%   | Gives a correct overshoot-and-correct mechanism AND a concrete, well-explained example (e.g. the cobweb cycle) that plausibly produces alternating-sign residuals. |
| Good        | 70–89%    | Correct mechanism, thinner or less concrete example.                                                                                                               |
| Partial     | 40–69%    | Vague reference to "some kind of cycle" without a worked example.                                                                                                  |
| Minimal     | 0–39%     | No credible mechanism given.                                                                                                                                       |

**(c)** For each of the following scenarios, state whether autocorrelation is present, based on the Durbin Watson test with a 5% level of significance. Briefly explain your answer.

**(i)**  $d = 1.56$ ,  $n = 25$ ,  $k = 3$

**(ii)**  $d = 2.89$ ,  $n = 30$ ,  $k = 5$

**(iii)**  $d = 2.35$ ,  $n = 20$ ,  $k = 4$

**(iv)**  $d = 1.55$ ,  $n = 40$ ,  $k = 2$

20%

### FULL MODEL ANSWER — PART (C)

The Durbin-Watson decision rule splits the 0–4 range into five zones using  $d_L$  and  $d_U$  from Table A5: positive autocorrelation if  $d < d_L$ ; inconclusive if  $d_L \leq d \leq d_U$ ; no autocorrelation if  $d_U < d < 4 - d_U$ ; inconclusive again if  $4 - d_U \leq d \leq 4 - d_L$ ; and negative autocorrelation if  $d > 4 - d_L$ .

#### Working — all four scenarios against Table A5

| Scenario                        | $d_L$ | $d_U$ | $4 - d_U$ | $4 - d_L$ | Zone                          | Verdict                  |
|---------------------------------|-------|-------|-----------|-----------|-------------------------------|--------------------------|
| (i) $d=1.56$ , $n=25$ , $k=3$   | 1.206 | 1.550 | 2.450     | 2.794     | $d_U < d < 4 - d_U$           | No autocorrelation       |
| (ii) $d=2.89$ , $n=30$ , $k=5$  | 1.143 | 1.739 | 2.261     | 2.857     | $d > 4 - d_L$                 | Negative autocorrelation |
| (iii) $d=2.35$ , $n=20$ , $k=4$ | 0.998 | 1.676 | 2.324     | 3.002     | $4 - d_U \leq d \leq 4 - d_L$ | Inconclusive             |
| (iv) $d=1.55$ , $n=40$ , $k=2$  | 1.442 | 1.544 | 2.456     | 2.558     | $d_U < d < 4 - d_U$           | No autocorrelation       |

**(i)**  $d = 1.56$  sits just above  $d_U = 1.550$ , placing it in the "no autocorrelation" zone — though only just; a slightly larger sample or a marginally different  $d$  could tip this into the inconclusive zone, so this result is directionally clear but not by a wide margin.

**(ii)**  $d = 2.89$  exceeds  $4 - d_L = 2.857$ , placing it in the negative-autocorrelation zone.

**(iii)**  $d = 2.35$  falls between  $4 - d_U = 2.324$  and  $4 - d_L = 3.002$  — the test is **inconclusive** here; we cannot reject or accept the null of no autocorrelation with confidence at this significance level.

**(iv)**  $d = 1.55$  sits just above  $d_U = 1.544$  — again "no autocorrelation," but by an even narrower margin than (i), essentially right on the boundary.

#### SOURCING NOTE

$d_L/d_U$  values are read directly, without interpolation, from Table A5 as reproduced in the exam paper's own appendix (all four  $(n,k)$  combinations are listed exactly).

**Mark Allocation — Part (c) (20 marks total, 5 marks per scenario)**

| Tier               | % of part | Criteria                                                                                                                                                                                                  |
|--------------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Full credit</b> | 90–100%   | Correctly reads $d_L/d_U$ for all four (n,k) pairs, correctly applies the five-zone rule (including both inconclusive zones), and reaches the right verdict for all four, with brief correct explanation. |
| <b>Good</b>        | 70–89%    | Three of four scenarios correct; understands the decision rule but makes one lookup or zone error.                                                                                                        |
| <b>Partial</b>     | 40–69%    | Only tests for positive autocorrelation ( $d < d_L$ ) and misses the negative-autocorrelation and inconclusive zones entirely.                                                                            |
| <b>Minimal</b>     | 0–39%     | Incorrect or missing $d_L/d_U$ values, or no defensible verdicts.                                                                                                                                         |

**(d)** Suppose a researcher wants to examine trends in smoking among teenagers over time, and collects quarterly data on a number of variables, proposing the following model:

$$SR_t = \beta_0 + \beta_1 INC_t + \beta_2 AS_t + \beta_3 SR_{t-1} + \beta_4 GDP_t + \varepsilon_t$$

where  $SR_t$  = smoking rate (%) among teenagers in quarter  $t$ ,  $INC_t$  = average weekly teenage income (€) in quarter  $t$ ,  $AS_t$  = cigarette advertising spend (€m) in quarter  $t$ ,  $GDP_t$  = Gross Domestic Product (€m) in quarter  $t$ . Given that this is a time series regression, the researcher believes it may suffer from autocorrelation and wants to test this formally. How would you recommend the researcher goes about this in this case? Justify your answer.

20%

### FULL MODEL ANSWER — PART (D)

#### VERDICT

Use the Breusch-Godfrey LM test, not the Durbin-Watson test.

**Why the Durbin-Watson test cannot be used here.** The regression includes  $SR_{t-1}$ , the **lagged dependent variable**, as a regressor. The Durbin-Watson test is known to be biased toward 2 (i.e. biased toward "no autocorrelation") whenever a lagged dependent variable appears among the regressors, because the lagged dependent variable is mechanically correlated with the very autocorrelation the test is trying to detect. Using DW here would risk failing to detect genuine autocorrelation that is actually present.

**Recommended test: the Breusch-Godfrey LM test.** This test is specifically designed to remain valid when regressors include a lagged dependent variable (or other non-strictly-exogenous regressors), and it can test for higher-order autocorrelation (not just first-order, unlike the basic DW test).

#### Steps.

1. Estimate the original model by OLS and obtain the residuals  $\hat{\varepsilon}_t$ .
2. Run the auxiliary regression of  $\hat{\varepsilon}_t$  on all the original regressors ( $INC_t$ ,  $AS_t$ ,  $SR_{t-1}$ ,  $GDP_t$ ) plus  $p$  lagged residuals  $\hat{\varepsilon}_{t-1}$ , ...,  $\hat{\varepsilon}_{t-p}$  ( $p = 1$  is standard for a first test; with quarterly data,  $p = 4$  is also worth testing to check for seasonal serial correlation).
3. Obtain  $R^2$  from this auxiliary regression and compute the test statistic  $(n-p)R^2$ , which is asymptotically  $\chi^2$  distributed with  $p$  degrees of freedom under  $H_0$ : no autocorrelation up to order  $p$ .
4. Reject  $H_0$  if the test statistic exceeds the critical  $\chi^2_p$  value at the chosen significance level, concluding that autocorrelation is present.

#### SOURCING NOTE

Breusch, T.S. (1978). "Testing for Autocorrelation in Dynamic Linear Models." *Australian Economic Papers*, 17, 334–355. Godfrey, L.G. (1978). "Testing Against General Autoregressive and Moving Average Error Models when the Regressors Include Lagged Dependent Variables." *Econometrica*, 46, 1293–1302. These are the two original papers behind the test, chosen specifically because their titles describe exactly this scenario — a regression with a lagged dependent variable.

**Mark Allocation — Part (d) (20 marks total)**

| Tier               | % of part | Criteria                                                                                                                                                                                                                       |
|--------------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Full credit</b> | 90–100%   | Correctly identifies that DW is invalid due to the lagged dependent variable, recommends the Breusch-Godfrey LM test, AND correctly describes its steps (auxiliary regression, $(n-p)R^2$ statistic, $\chi^2_p$ distribution). |
| <b>Good</b>        | 70–89%    | Correctly recommends Breusch-Godfrey and explains why DW fails, but the test steps are incomplete.                                                                                                                             |
| <b>Partial</b>     | 40–69%    | Recommends DW without noticing the lagged-dependent-variable problem, or names Breusch-Godfrey without justification.                                                                                                          |
| <b>Minimal</b>     | 0–39%     | No credible test recommended.                                                                                                                                                                                                  |

**Question A3 — Total: 100%**